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SCHOOL OF COMMUNICATION AND COMPUTER STUDIES
BACHELOR OF SCIENCE IN COMPUTER SCIENCE
JANUARY - APRIL 2024/2025 SEMESTER
CSC 4208 : NUMERICAL COMPUTATION

DATE: APRIL, 2025

TIME: 2 Hours

INSTRUCTIONS

Section A is compulsory. Answer any other two questions in section B

SECTION A(COMPULSORY - This section carries 30 Marks)

Question One

- (a) Differentiate between the following, absolute error and relative error (2 Marks) — 2
- (b) Explain two types of errors (4 Marks) —
- (c) Two approximate numbers x_1 and x_2 are given correct to their last digit, $x_1 = 5.16$ and $x_2 = 1.2$
Find the relative error and absolute in computing x_1/x_2 (5 Marks) —
- (d) Find the approximation of $f(1.14)$ given the following function values

x_i	f_i
1.1	0.8912
1.2	0.9320

(4 Marks)

(e) Solve the system of equations using LU decomposition method

$$2x + 3y + z = 9$$

$$x + 2y + 3z = 6$$

$$3x + y + 2z = 8$$

(6 Marks) - 4

(f) Describe two methods of curve fitting in numerical computation

(4 Marks) - 4

(g) State four basic quadrature rules in numerical integration

(4 Marks) - 4

(h) Differentiate the following function with respect to x

$$f(x) = 4x^3 + 2x - 6x^4 + 2$$

(1 Mark)

SECTION B (Answer ANY TWO. Each question carries 20 Marks)

Question Two

(a) The efficiency of solar energy collectors S_1 and S_2 computed using the formula

$$\eta = K \frac{QT_d}{I}$$

were found to be 0.76 and 0.70 respectively. The errors committed in measuring Q , T_d and I are given in the table below

Collector	S_1	S_2
Q	1.5%	0.5%
T_d	1%	1%
I	3.6%	2%

Determine which best collector is the best

(6 Marks)

(b) Solve the following system of equation by Gauss-Elimination method

$$2x + 2y + 4z = 15$$

$$x + 3y + 2z = 13$$

$$3x + y + 3z = 14$$

(9 Marks)

(c) Compute $\frac{dy}{dx}$ for $x = 1$

x	1	2	3	4	5	6
y	1	8	27	64	125	216

(5 Marks)

Question Three

(a) Calculate the values of x , y and z in the system below using Gauss Jacobi's method (perform 4 iterations)

$$5x + 2y + z = 12$$

$$x + 4y + 2z = 15$$

$$x + 2y + 5z = 20$$

(7 Marks) - 7

(b) Fit a least square line $y = a + bx$ to the data

x	-2	-1	0	1	2
y	1	2	3	3	4

(8 Marks) - 8 / 15

(c) Write python code that estimate and print the fitted line in (b)

(5 Marks)

Question Four

(a) Calculate the value of $\int_0^1 \frac{x}{1+x} dx$ taking six intervals by Trapezoidal rule

(5 Marks) - 5

(b) Given $\frac{dy}{dx} = 1 + xy$ with the initial condition that $y = 0$ at $x = 0$. Compute $y(0.1)$

(6 Marks) - 6

(c) Find a roots of the equation $f(x) = x^3 - x - 1$ using fixed point iteration method (in 5 iterations)

(9 Marks) - 4 / 15

Question Five

(a) Solve the equation $\frac{dy}{dx} = 1 - y$ with the initial condition $x = 0, y = 0$ using Euler's algorithm and tabulate solutions at $x = 0.1, 0.2, 0.3$

(9 Marks)

(b) Write a python code to compute $y(0.2)$ in (a) by Euler's method

(7 Marks)

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$$\frac{15}{26} \frac{26}{52}$$

$$\frac{30}{26} \frac{26}{56}$$

$$\frac{25}{54} \frac{54}{79}$$

$$\frac{25}{54} \frac{54}{76}$$

(c) Explain the following

(2 Marks)

- i) Adaptive quadrature
- ii) Difference between single-step and multi-step methods in solving ordinary differential equations

(2 Marks)